

Nilaa Raghunathan

(943) 238-5010 | nr3005@columbia.edu | [linkedin.com/in/nilaaraghunathan/](https://www.linkedin.com/in/nilaaraghunathan/) | github.com/Nilaa-Jaya

EDUCATION

Columbia University

MS in Data Science

New York, US

Sep 2024 - Dec 2025

- Coursework: Generative AI, Deep Learning, Machine Learning, Data Structures & Algorithms, Probability & Statistics.

Vellore Institute of Technology

BTech in Computer Science with Specialization in Data Science

Vellore, IN

Sep 2020 - May 2024

- Coursework: NLP, Artificial Intelligence, Business Intelligence, Cloud Computing, Data Analytics, Software Engineering.

TECHNICAL SKILLS

Languages: Python, SQL, Scala, C, C#, Java, TypeScript.

Distributed Systems: Apache Spark, PySpark, Kafka, Airflow, Databricks, Synapse, PostgreSQL, MySQL, Redis, Delta Lake.

Cloud & DevOps: Azure, AWS, Docker, Kubernetes, Git, Github Copilot, CI/CD, Jenkins, Linux, Azure Data Explorer (Kusto).

AI/ML: PyTorch, TensorFlow, scikit-learn, LLMs, RAG, Hugging Face, LangChain, FastAPI, PEFT, MLflow, Azure AI Foundry

EXPERIENCE

Limex Trading

Applied AI Intern

New York, US

Sep 2025 - Dec 2025

- Built an agentic financial research copilot combining Hybrid RAG, GraphRAG, LangGraph orchestration, and multi-hop knowledge retrieval, reducing analyst research time by 42%.
- Improved retrieval precision by 24% & Recall@10 by 18% using vector search, BM25, graph traversal & cross-encoder reranking.
- Engineered a financial knowledge graph with 35K+ nodes & 120K+ relationships, using schema-constrained extraction and confidence-scored validation for data quality and traceability.
- Fine-tuned Llama 3.1-8B with QLoRA on 600+ financial instruction pairs, improving citation accuracy from 72% to 81%.

Magna International Inc.

Data Science Intern

Michigan, US

Jun 2025 - Aug 2025

- Built an ML-based inventory forecasting pipeline using Azure Synapse and Databricks across 1,700+ automotive parts, reducing safety stock by 12% and saving \$1.4M annually.
- Increased forecast accuracy by 18% (MAPE) using probabilistic time-series modeling (GMM-HMM) to capture demand patterns.
- Strengthened demand signal reliability by building a context-aware NLP pipeline (BiLSTM-RoBERTa) over 7K+ supplier records.
- Lowered stock-out risk by 13% through ANOVA, A/B testing & causal impact analysis, validating changes before deployment.
- Deployed a Streamlit application backed by Databricks and MySQL, enabling planners to monitor forecasts and inventory risk.

Vellore Institute of Technology

Applied Research Intern

Vellore, IN

Apr 2022 - Jul 2023

- Architected a distributed data platform using Kafka and Spark for scalable ingestion and transformation of 50M+ records.
- Built automated Airflow ETL pipelines with schema validation, data quality checks and monitoring to improve pipeline reliability.
- Published findings as first-author IEEE Access research on large-scale sentiment analysis across multilingual, cross-domain, and low-resource datasets, receiving 85+ citations.

PROJECTS

Multi-Agent HR Intelligence Platform (github.com/Nilaa-Jaya/Multi-Agent-HR-Intelligence-Platform-)

May 2026 - Present

- Architected and deployed multi-agent HR support system using LangGraph/LangChain with 9 specialized agents, implementing 7-way conditional routing and stateful orchestration to automate 85% of HR inquiries with 90%+ resolution accuracy.
- Built scalable RAG infrastructure with FAISS vector search and re-ranking, reducing query handling time by 30% and errors by 40% while maintaining <1s retrieval latency.
- Developed production FastAPI microservices with PostgreSQL analytics, Docker deployment, GitHub Actions CI/CD (38 tests), and secure webhooks for reliable system integration.

Generative Recommender System (github.com/Nilaa-Jaya/Generative-Recommender-System)

Feb 2026 - May 2026

- Designed RAG-based personalized-recommender system processing 230M+ reviews, 54M+ users & 1.57M+ products.
- Built scalable vector search system using Sentence-BERT, FAISS, diversity re-ranking, and persona-based retrieval via DeepMF embeddings, improving Precision@10 from 0.57 to 0.69 and Diversity@10 from 0.42 to 0.51 with ~43ms latency.
- Fine-tuned LLaMA2-7B using QLoRA with RLHF reward model on 100K pairs, improving BERTScore from 0.79 to 0.86

Financial Fraud Detection (github.com/Nilaa-Jaya/Financial-Fraud-Detection)

Aug 2025 - Oct 2025

- Developed fraud detection pipeline on 6.3M+ transactions, engineering error-based and anomaly features using Isolation Forest.
- Boosted fraud detection by ADASYN oversampling with ML (Logistic Regression, Random Forest, XGBoost) & deep learning models (RNN, LSTM, BiLSTM) achieving 0.98 F1-score, 97% recall & reducing false negatives by 50%.

PUBLICATIONS

- Challenges and Issues in Sentiment Analysis: Comprehensive Survey | *IEEE Access* (Cited by 85+).
- Improved Deep Learning Model for Sentiment Analysis using LSTM with Layer Normalization | *IEEE - AAIAC Conference*.